Volumes by the Disc and Washer Methods

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: The Disc Method Around Coordinate Axes**

The base of a solid is the region under $y = 3\sqrt{x}$ on $[0, 11]$. Cross-sections perpendicular to the x-axis are squares with side length y. Find the volume of the solid.

Final Answer: $\frac{1089}{2}$. Each cross-section has area $y^2 = (3\sqrt{x})^2 = 9x$. $V = \int_0^{11} 9x\,dx = \frac{1089}{2}$.

2 **[Foundational]** • **Target: The Washer Method Around Coordinate Axes**

The base of a solid is the region under $y = 1\sqrt{x}$ on $[0, 11]$. Cross-sections perpendicular to the x-axis are squares with side length y. Find the volume of the solid.

Final Answer: $\frac{121}{2}$. Each cross-section has area $y^2 = (1\sqrt{x})^2 = 1x$. $V = \int_0^{11} 1x\,dx = \frac{121}{2}$.

3 **[Foundational]** • **Target: Revolutions Around Non-Axis Horizontal Lines**

The base of a solid is the region under $y = 3\sqrt{x}$ on $[0, 7]$. Cross-sections perpendicular to the x-axis are squares with side length y. Find the volume of the solid.

Final Answer: $\frac{441}{2}$. Each cross-section has area $y^2 = (3\sqrt{x})^2 = 9x$. $V = \int_0^{7} 9x\,dx = \frac{441}{2}$.

4 **[Foundational]** • **Target: Volumes with Known Cross-Sections**

The base of a solid is the region under $y = 3\sqrt{x}$ on $[0, 9]$. Cross-sections perpendicular to the x-axis are squares with side length y. Find the volume of the solid.

Final Answer: $\frac{729}{2}$. Each cross-section has area $y^2 = (3\sqrt{x})^2 = 9x$. $V = \int_0^{9} 9x\,dx = \frac{729}{2}$.

5 **[Foundational]** • **Target: The Disc Method Around Coordinate Axes**

The base of a solid is the region under $y = 1\sqrt{x}$ on $[0, 5]$. Cross-sections perpendicular to the x-axis are squares with side length y. Find the volume of the solid.

Final Answer: $\frac{25}{2}$. Each cross-section has area $y^2 = (1\sqrt{x})^2 = 1x$. $V = \int_0^{5} 1x\,dx = \frac{25}{2}$.

6 **[Foundational]** • **Target: The Washer Method Around Coordinate Axes**

The base of a solid is the region under $y = 3\sqrt{x}$ on $[0, 3]$. Cross-sections perpendicular to the x-axis are squares with side length y. Find the volume of the solid.

Final Answer: $\frac{81}{2}$. Each cross-section has area $y^2 = (3\sqrt{x})^2 = 9x$. $V = \int_0^{3} 9x\,dx = \frac{81}{2}$.

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7 [Foundational] • Target: Revolutions Around Non-Axis Horizontal Lines

The base of a solid is the region under $y = 1\sqrt{x}$ on $[0, 9]$. Cross-sections perpendicular to the x-axis are squares with side length y. Find the volume of the solid.

Final Answer: $\frac{81}{2}$. Each cross-section has area $y^2 = (1\sqrt{x})^2 = 1x$. $V = \int_0^9 1x\,dx = \frac{81}{2}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Volumes with Known Cross-Sections

The base of a solid is the region under $y = 4\sqrt{x}$ on $[0, 10]$. Cross-sections perpendicular to the x-axis are squares with side length y. Find the volume of the solid.

Final Answer: 800. Each cross-section has area $y^2 = (4\sqrt{x})^2 = 16x$. $V = \int_0^{10} 16x\,dx = 800$.

9 [Intermediate] • Target: The Disc Method Around Coordinate Axes

The base of a solid is the region under $y = 4\sqrt{x}$ on $[0, 4]$. Cross-sections perpendicular to the x-axis are squares with side length y. Find the volume of the solid.

Final Answer: 128. Each cross-section has area $y^2 = (4\sqrt{x})^2 = 16x$. $V = \int_0^4 16x\,dx = 128$.

10 [Intermediate] • Target: The Washer Method Around Coordinate Axes

The base of a solid is the region under $y = 4\sqrt{x}$ on $[0, 8]$. Cross-sections perpendicular to the x-axis are squares with side length y. Find the volume of the solid.

Final Answer: 512. Each cross-section has area $y^2 = (4\sqrt{x})^2 = 16x$. $V = \int_0^8 16x\,dx = 512$.

11 [Intermediate] • Target: Revolutions Around Non-Axis Horizontal Lines

The base of a solid is the region under $y = 4\sqrt{x}$ on $[0, 9]$. Cross-sections perpendicular to the x-axis are squares with side length y. Find the volume of the solid.

Final Answer: 648. Each cross-section has area $y^2 = (4\sqrt{x})^2 = 16x$. $V = \int_0^9 16x\,dx = 648$.

12 [Intermediate] • Target: Volumes with Known Cross-Sections

The base of a solid is the region under $y = 4\sqrt{x}$ on $[0, 7]$. Cross-sections perpendicular to the x-axis are squares with side length y. Find the volume of the solid.

Final Answer: 392. Each cross-section has area $y^2 = (4\sqrt{x})^2 = 16x$. $V = \int_0^7 16x\,dx = 392$.

13 [Intermediate] • Target: The Disc Method Around Coordinate Axes

The base of a solid is the region under $y = 2\sqrt{x}$ on $[0, 11]$. Cross-sections perpendicular to the x-axis are squares with side length y. Find the volume of the solid.

Final Answer: 242. Each cross-section has area $y^2 = (2\sqrt{x})^2 = 4x$. $V = \int_0^{11} 4x\,dx = 242$.

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14 [Intermediate] • Target: The Washer Method Around Coordinate Axes

The base of a solid is the region under $y = 2\sqrt{x}$ on $[0, 10]$. Cross-sections perpendicular to the x-axis are squares with side length y. Find the volume of the solid.

Final Answer: 200. Each cross-section has area $y^2 = (2\sqrt{x})^2 = 4x$. $V = \int_0^{10} 4x\,dx = 200$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Revolutions Around Non-Axis Horizontal Lines • Cross-Lesson: 1.01 Limits Graphically and Numerically

The base of a solid is the region under $y = 4\sqrt{x}$ on $[0, 3]$. Cross-sections perpendicular to the x-axis are squares with side length y. Find the volume of the solid.

Final Answer: 72. Each cross-section has area $y^2 = (4\sqrt{x})^2 = 16x$. $V = \int_0^{3} 16x\,dx = 72$.

16 [Challenging] • Target: Volumes with Known Cross-Sections • Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem

The base of a solid is the region under $y = 3\sqrt{x}$ on $[0, 8]$. Cross-sections perpendicular to the x-axis are squares with side length y. Find the volume of the solid.

Final Answer: 288. Each cross-section has area $y^2 = (3\sqrt{x})^2 = 9x$. $V = \int_0^{8} 9x\,dx = 288$.

17 [Challenging] • Target: The Disc Method Around Coordinate Axes • Cross-Lesson: 2.01 The Definition of the Derivative

The base of a solid is the region under $y = 2\sqrt{x}$ on $[0, 8]$. Cross-sections perpendicular to the x-axis are squares with side length y. Find the volume of the solid.

Final Answer: 128. Each cross-section has area $y^2 = (2\sqrt{x})^2 = 4x$. $V = \int_0^{8} 4x\,dx = 128$.

18 [Challenging] • Target: The Washer Method Around Coordinate Axes • Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives

The base of a solid is the region under $y = 3\sqrt{x}$ on $[0, 6]$. Cross-sections perpendicular to the x-axis are squares with side length y. Find the volume of the solid.

Final Answer: 162. Each cross-section has area $y^2 = (3\sqrt{x})^2 = 9x$. $V = \int_0^{6} 9x\,dx = 162$.

19 [Challenging] • Target: Revolutions Around Non-Axis Horizontal Lines • Cross-Lesson: 3.01 The Chain Rule

The base of a solid is the region under $y = 3\sqrt{x}$ on $[0, 4]$. Cross-sections perpendicular to the x-axis are squares with side length y. Find the volume of the solid.

Final Answer: 72. Each cross-section has area $y^2 = (3\sqrt{x})^2 = 9x$. $V = \int_0^{4} 9x\,dx = 72$.

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20 [Challenging] • Target: Volumes with Known Cross-Sections • Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions

The base of a solid is the region under $y = 2\sqrt{x}$ on $[0, 4]$. Cross-sections perpendicular to the x-axis are squares with side length y. Find the volume of the solid.

Final Answer: 32. Each cross-section has area $y^2 = (2\sqrt{x})^2 = 4x$. $V = \int_0^4 4x\,dx = 32$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: The Disc Method Around Coordinate Axes

Find the volume of the solid generated by revolving the region bounded by $y = \sqrt{x}$, the x-axis, and $x = 4$ about the x-axis.

a. 8π ✓

b. 16π

c. 4π

d. $\frac{16\pi}{3}$

Disc formula: $V = \pi\int_0^4 [R(x)]^2 dx = \pi\int_0^4 (\sqrt{x})^2 dx = \pi\int_0^4 x dx = \pi[\frac{x^2}{2}]_0^4 = 8\pi$.

22 [Foundational] • Target: The Disc Method Around Coordinate Axes

Find the volume when the region bounded by $y = x^2$, $x = 0$, and $y = 4$ is revolved about the y-axis.

a. 8π ✓

b. 16π

c. 4π

d. $\frac{32\pi}{5}$

Revolving about y-axis: $x = \sqrt{y}$. $V = \pi\int_0^4 (\sqrt{y})^2 dy = \pi\int_0^4 y dy = \pi[\frac{y^2}{2}]_0^4 = 8\pi$.

23 [Foundational] • Target: The Disc Method Around Coordinate Axes

What is the cross-sectional shape of a solid formed by the disc method?

a. **A solid circular disk of area** πR^2 ✓

b. A circular ring (washer)

c. A rectangle

d. A cylindrical shell

The disc method generates perpendicular slices that are solid circular disks with area $A(x) = \pi[R(x)]^2$.

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24 [Foundational] • Target: The Washer Method Around Coordinate Axes

The region enclosed by $y = x$ and $y = x^2$ is revolved about the x-axis. What is the resulting volume?

a. $\frac{2\pi}{15}$ ✓

b. $\frac{\pi}{6}$

c. $\frac{\pi}{15}$

d. $\frac{4\pi}{15}$

Intersections at $x = 0, 1$. Outer radius $R(x) = x$, inner radius $r(x) = x^2$.
$V = \pi\int_0^1 (x^2 - x^4)dx = \pi[\frac{x^3}{3} - \frac{x^5}{5}]_0^1 = \pi(\frac{1}{3} - \frac{1}{5}) = \frac{2\pi}{15}$.

25 [Foundational] • Target: The Washer Method Around Coordinate Axes

Write the integral for the volume when the region between $y = 2x$ and $y = x^2$ is revolved about the x-axis.

a. $\pi\int_0^2 (4x^2 - x^4)dx$ ✓

b. $\pi\int_0^2 (2x - x^2)^2 dx$

c. $\pi\int_0^2 (x^4 - 4x^2)dx$

d. $2\pi\int_0^2 (2x - x^2)dx$

Outer radius is $2x$, inner radius is x^2. Washer area is $\pi[(2x)^2 - (x^2)^2] = \pi(4x^2 - x^4)$.

26 [Foundational] • Target: The Washer Method Around Coordinate Axes

Why is $\pi\int (R - r)^2 dx$ INCORRECT for the washer method?

a. **The area of a washer is $\pi R^2 - \pi r^2$, which is NOT equal to $\pi(R - r)^2$** ✓

b. It omits the factor of 2

c. It is only valid for shells

d. It produces negative volume

Area of an annulus is $\pi(R^2 - r^2) = \pi R^2 - \pi r^2$. Squaring the difference $(R - r)^2 = R^2 - 2Rr + r^2$ introduces an erroneous $-2Rr$ term.

27 [Foundational] • Target: Revolutions Around Non-Axis Horizontal Lines

The region bounded by $y = x^2$ and $y = 4$ is revolved about the line $y = 4$. What is the outer radius $R(x)$?

a. $R(x) = 4 - x^2$ ✓

b. $R(x) = x^2$

c. $R(x) = 4$

d. $R(x) = x^2 - 4$

The axis of revolution is $y = 4$. The distance from the curve $y = x^2$ to the axis $y = 4$ is $4 - x^2$, which forms solid disks with radius $4 - x^2$.

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28 [Foundational] • Target: Revolutions Around Non-Axis Horizontal Lines

Find the volume in the above problem revolved about $y = 4$.

a. $\frac{256\pi}{15}$ ✓

b. $\frac{128\pi}{15}$

c. $\frac{64\pi}{5}$

d. 16π

$V = \pi\int_{-2}^{2}(4-x^2)^2dx = 2\pi\int_{0}^{2}(16-8x^2+x^4)dx = 2\pi[16x - \frac{8x^3}{3} + \frac{x^5}{5}]_0^2 = 2\pi[32 - \frac{64}{3} + \frac{32}{5}] = 2\pi[\frac{480-320+96}{15}] = 2\pi$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Revolutions Around Non-Axis Horizontal Lines

The region between $y = \sqrt{x}$, $y = 0$, and $x = 4$ is revolved about the line $y = -1$. What are the washer radii?

a. $R(x) = \sqrt{x} + 1$ **and** $r(x) = 1$ ✓

b. $R(x) = \sqrt{x} - 1$ and $r(x) = -1$

c. $R(x) = \sqrt{x}$ and $r(x) = 1$

d. $R(x) = 4$ and $r(x) = \sqrt{x}$

Outer curve is $y = \sqrt{x}$, distance to $y = -1$ is $\sqrt{x} - (-1) = \sqrt{x} + 1$. Inner boundary is $y = 0$, distance to $y = -1$ is $0 - (-1) = 1$.

30 [Intermediate] • Target: Volumes with Known Cross-Sections

The base of a solid is the region enclosed by $y = 1 - x^2$ and the x-axis. Cross-sections perpendicular to the x-axis are squares. What is the volume?

a. $\frac{16}{15}$ ✓

b. $\frac{8}{15}$

c. $\frac{4}{3}$

d. $\frac{32}{15}$

Side of square is $s(x) = 1 - x^2$. Area $A(x) = (1-x^2)^2 = 1 - 2x^2 + x^4$.
$V = \int_{-1}^{1}(1-2x^2+x^4)dx = 2[x - \frac{2x^3}{3} + \frac{x^5}{5}]_0^1 = 2(1 - \frac{2}{3} + \frac{1}{5}) = 2(\frac{8}{15}) = \frac{16}{15}$.

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31 [Intermediate] • Target: Volumes with Known Cross-Sections

If cross-sections perpendicular to the x-axis are semicircles with diameter on the base, what is the cross-sectional area formula?

a. $A(x) = \frac{\pi}{8}[s(x)]^2$ ✓

b. $A(x) = \frac{\pi}{2}[s(x)]^2$

c. $A(x) = \frac{\pi}{4}[s(x)]^2$

d. $A(x) = \pi[s(x)]^2$

Radius is $r = \frac{s(x)}{2}$. Semicircle area is $\frac{1}{2}\pi r^2 = \frac{1}{2}\pi(\frac{s}{2})^2 = \frac{\pi}{8}s^2$.

32 [Intermediate] • Target: Volumes with Known Cross-Sections

If cross-sections are equilateral triangles with side length $s(x)$, what is the cross-sectional area formula?

a. $A(x) = \frac{\sqrt{3}}{4}[s(x)]^2$ ✓

b. $A(x) = \frac{1}{2}[s(x)]^2$

c. $A(x) = \frac{\sqrt{3}}{2}[s(x)]^2$

d. $A(x) = \frac{1}{4}[s(x)]^2$

The area of an equilateral triangle with side s is $\frac{\sqrt{3}}{4}s^2$.

33 [Intermediate] • Target: The Disc Method Around Coordinate Axes

The base of a solid is the region under $y = 1\sqrt{x}$ on $[0, 8]$. Cross-sections perpendicular to the x-axis are squares with side length y. Find the volume of the solid.

a. 64

b. 30

c. **32** ✓

d. 34

Each cross-section has area $y^2 = (1\sqrt{x})^2 = 1x$. $V = \int_0^8 1x\,dx = 32$.

34 [Intermediate] • Target: The Washer Method Around Coordinate Axes

The base of a solid is the region under $y = 2\sqrt{x}$ on $[0, 7]$. Cross-sections perpendicular to the x-axis are squares with side length y. Find the volume of the solid.

a. 196

b. 100

c. 96

d. **98** ✓

Each cross-section has area $y^2 = (2\sqrt{x})^2 = 4x$. $V = \int_0^7 4x\,dx = 98$.

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35 [Intermediate] • Target: Revolutions Around Non-Axis Horizontal Lines

The base of a solid is the region under $y = 1\sqrt{x}$ on $[0, 4]$. Cross-sections perpendicular to the x-axis are squares with side length y. Find the volume of the solid.

a. 6

b. **8** ✓

c. 10

d. 16

Each cross-section has area $y^2 = (1\sqrt{x})^2 = 1x$. $V = \int_0^4 1x\,dx = 8$.

36 [Intermediate] • Target: Volumes with Known Cross-Sections

The region between $y = 8$ and $y = x^2$ on $[0, 2]$ is rotated about the x-axis. Find the volume using the washer method.

a. $\frac{608}{5}$

b. $\frac{304}{5}\pi$

c. $\frac{1216}{5}\pi$

d. $\frac{608}{5}\pi$ ✓

Washer method: $V = \pi\int_0^2 (8^2 - (x^2)^2)dx = \frac{608}{5}\pi$.

37 [Intermediate] • Target: The Disc Method Around Coordinate Axes

The region between $y = 9$ and $y = x^2$ on $[0, 2]$ is rotated about the x-axis. Find the volume using the washer method.

a. $\frac{778}{5}$

b. $\frac{389}{5}\pi$

c. $\frac{1556}{5}\pi$

d. $\frac{778}{5}\pi$ ✓

Washer method: $V = \pi\int_0^2 (9^2 - (x^2)^2)dx = \frac{778}{5}\pi$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: The Washer Method Around Coordinate Axes • Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives

The region between $y = 9$ and $y = x^2$ on $[0, 1]$ is rotated about the x-axis. Find the volume using the washer method.

a. $\frac{202}{5}\pi$

b. $\frac{808}{5}\pi$

c. $\frac{404}{5}\pi$ ✓

d. $\frac{404}{5}$

Washer method: $V = \pi\int_0^1 (9^2 - (x^2)^2)dx = \frac{404}{5}\pi$.

39 [Challenging] • Target: Revolutions Around Non-Axis Horizontal Lines • Cross-Lesson: 3.01 The Chain Rule

The region between $y = 6$ and $y = x^2$ on $[0, 1]$ is rotated about the x-axis. Find the volume using the washer method.

a. $\frac{179}{10}\pi$

b. $\frac{179}{5}$

c. $\frac{358}{5}\pi$

d. $\frac{179}{5}\pi$ ✓

Washer method: $V = \pi\int_0^1 (6^2 - (x^2)^2)dx = \frac{179}{5}\pi$.

40 [Challenging] • Target: Volumes with Known Cross-Sections • Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions

The region between $y = 2$ and $y = x^2$ on $[0, 1]$ is rotated about the x-axis. Find the volume using the washer method.

a. $\frac{19}{5}$

b. $\frac{19}{10}\pi$

c. $\frac{19}{5}\pi$ ✓

d. $\frac{38}{5}\pi$

Washer method: $V = \pi\int_0^1 (2^2 - (x^2)^2)dx = \frac{19}{5}\pi$.

41 [Challenging] • Target: The Disc Method Around Coordinate Axes • Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion

The region between $y = 3$ and $y = x^2$ on $[0, 1]$ is rotated about the x-axis. Find the volume using the washer method.

a. $\frac{44}{5}$

b. $\frac{22}{5}\pi$

c. $\frac{88}{5}\pi$

d. $\frac{44}{5}\pi$ ✓

Washer method: $V = \pi\int_0^1 (3^2 - (x^2)^2)dx = \frac{44}{5}\pi$.

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42 [Challenging] • Target: The Washer Method Around Coordinate Axes • Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule

Consider the function $f(x)$ whose graph is shown below. The graph consists of a line segment and a quarter circle. The line segment connects the points $(2, 4)$ and $(4, 0)$. The quarter circle is in the second quadrant and is centered at $(0, 0)$ with radius 2. Let $A(x)$ be the area of the region bounded by the graph of $f(t)$, the t-axis, and the lines $t = 0$ and $t = x$. (a) Find $A(3)$.

a. **4** ✓

b. 6

c. 5

d. 7

The function $f(x)$ is defined piecewise. For $0 \le t \le 2$, $f(t)$ is the upper part of a circle centered at the origin with radius 2, so $f(t) = \sqrt{4 - t^2}$. For $2 < t \le 4$, $f(t)$ is a line segment connecting $(2, 4)$ and $(4, 0)$. The equation of this line can be found using the two-point form: $y - 0 = \frac{4-0}{2-4}(x - 4)$, which simplifies to $y = -2(x - 4)$, so $f(t) = -2(t - 4)$. To find $A(3)$, we need to calculate the definite integral $\int_0^3 f(t)\,dt$. We split this integral into two parts based on the definition of $f(t)$: $A(3) = \int_0^2 f(t)\,dt + \int_2^3 f(t)\,dt$ For the first part, $\int_0^2 \sqrt{4 - t^2}\,dt$, this represents the area of a quarter circle of radius 2, which is $\frac{1}{4}\pi r^2 = \frac{1}{4}\pi(2^2) = \pi$. For the second part, $\int_2^3 (-2(t - 4))\,dt$, we can evaluate this integral: $\int_2^3 (-2t + 8)\,dt = [-t^2 + 8t]_2^3$ Evaluating at the limits: $(-3^2 + 8 \cdot 3) - (-2^2 + 8 \cdot 2)$ $(-9 + 24) - (-4 + 16)$ $15 - 12 = 3$. Therefore, $A(3) = \pi + 3$. *Common Misconceptions:* Students might incorrectly assume the function is a single simple shape or miscalculate the line segment's equation. They might also forget that the first part of the integral represents a quarter circle's area and attempt to integrate $\sqrt{4 - t^2}$ directly, which is more complex.

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43 [Challenging] • Target: Revolutions Around Non-Axis Horizontal Lines • Cross-Lesson: 5.01 Extrema and the Mean Value Theorem

The graph of a differentiable function f is shown in the figure above. The function f has a local maximum at $x = c$ and a local minimum at $x = e$. The point $(c, f(c))$ is approximately $(1, 4)$ and the point $(e, f(e))$ is approximately $(4, -2)$. The shaded region, bounded by the graph of f, the x-axis, and the lines $x = 0$ and $x = 5$, has area 12. $If\int_{0}^{c} f(x) \, dx = k, what is the value of \int_{c}^{5} f(x) \, dx in terms of k$?

a. 12 + k

b. **12 - k ✓**

c. k - 12

d. k + 2

The total area of the shaded region from $x = 0$ to $x = 5$ is given as 12. This total area can be expressed as the sum of the areas of two sub-regions: the area from $x = 0$ to $x = c$ and the area from $x = c$ to $x = 5$. Mathematically, this can be written using definite integrals: $\int_0^5 f(x)\,dx = \int_0^c f(x)\,dx + \int_c^5 f(x)\,dx$. We are given that the total area is 12, so $\int_0^5 f(x)\,dx = 12$. We are also given that $\int_0^c f(x)\,dx = k$. Substituting these values into the equation: $12 = k + \int_c^5 f(x)\,dx$. To find the value of $\int_c^5 f(x)\,dx$, we can rearrange the equation: $\int_c^5 f(x)\,dx = 12 - k$. Therefore, the value of $\int_c^5 f(x)\,dx$ in terms of k is $12 - k$.

44 [Challenging] • Target: Volumes with Known Cross-Sections • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

The graph of a function $f'(x)$ is shown below. It is given that $f(x)$ is a function such that $f(1) = 5$. The area between the curve $y = f'(x)$, the x-axis, and the vertical lines $x = 1$ and $x = 5$ is represented by the shaded region. If the area of the shaded region is 16 square units, determine the value of $f(5)$.

a. 17

b. **21 ✓**

c. 9

d. 11

The area of the shaded region is given by the definite integral of $f'(x)$ from $x = 1$ to $x = 5$. This area is also equal to the change in the function $f(x)$ over this interval: $\int_1^5 f'(x)\,dx = f(5) - f(1)$. The graph shows that the function $f'(x)$ is a constant function, represented by the horizontal line $y = 4$. Therefore, $f'(x) = 4$. The area of the shaded region is given as 16 square units. Using the definite integral: $\int_1^5 4\,dx = [4x]_1^5 = 4(5) - 4(1) = 20 - 4 = 16$. This confirms the given area. Now we use the relationship $f(5) - f(1) = 16$. We are given that $f(1) = 5$. Substituting this value, we get $f(5) - 5 = 16$. Solving for $f(5)$: $f(5) = 16 + 5 = 21$.

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45 **[Challenging]** • **Target: The Disc Method Around Coordinate Axes** • **Cross-Lesson: 6.01 Antiderivatives and Riemann Sums**

Consider the region bounded by the graph of $f(x) = x^2 + 2$, the x-axis, and the vertical lines $x = 0$ and $x = 3$. Which of the following represents the area of this region?

a. $\int_0^3 (x^2 + 2)\,dx$ ✓

b. $\int_0^3 (x^2 - 2)\,dx$

c. $\int_0^3 x^2\,dx$

d. $\int_0^3 2\,dx$

The area of a region bounded by a curve $y = f(x)$, the x-axis, and the vertical lines $x = a$ and $x = b$ is given by the definite integral $\int_a^b f(x)\,dx$, provided that $f(x) \geq 0$ over the interval [a, b]. In this case, $f(x) = x^2 + 2$, which is always non-negative for all real x. The region is bounded by $x = 0$ and $x = 3$. Therefore, the area of the region is represented by the definite integral $\int_0^3 (x^2 + 2)\,dx$.

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